

$$\begin{bmatrix} 0 & \textcircled{1} & 0 & 5 \\ 0 & 0 & \textcircled{1} & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

ref ✓
rref ✓

Linear system

$$\begin{aligned} X - 2y + 3z &= 9 \\ -X + 3y &= -4 \\ 2x - 5y + 5z &= 17 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{array} \right]$$

solve the systems

Part 3

$$x_2 + x_3 - 2x_4 = -3$$

$$x_1 + 2x_2 - x_3 = 2$$

$$2x_1 + 4x_2 + x_3 - 3x_4 = -2$$

$$x_1 - 4x_2 - 7x_3 - x_4 = -19$$

$$\left[\begin{array}{cccc|c} 0 & 1 & 1 & -2 & -3 \\ 1 & 2 & -1 & 0 & 2 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & -1 & -19 \end{array} \right]$$

$R_1 \leftrightarrow R_2$

$$\left[\begin{array}{cccc|c} \textcircled{1} & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & -1 & -19 \end{array} \right]$$

Part 7

Homogenous system

كل المتغيرات = 0

$$\begin{aligned} x_1 + 3x_2 - x_3 + x_4 &= 0 \\ x_1 + 2x_2 - 2x_4 &= 0 \\ 3x_1 + x_2 + 4x_3 &= 0 \end{aligned}$$

$$\left[\begin{array}{cccc|c} 1 & 3 & -1 & 1 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 3 & 1 & 4 & 0 & 0 \end{array} \right]$$

$$\left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{array} \right]$$

solve the homogenous
 $AX = 0$
 Find the specific nonzero solution

$$\left[\begin{array}{cccc|c} \textcircled{1} & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 \end{array} \right]$$

$$\begin{aligned} R_2 &= -R_1 + R_2 \\ R_3 &= -R_1 + R_3 \end{aligned}$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \end{array} \right]$$

$$A^{-1} = \begin{pmatrix} -2 & -5 & -1 \\ -3 & -5 & -1 \\ -2 & -4 & -1 \end{pmatrix}$$

لویايز تتاكرو صا اظرب المصفوفة ده

X او مصفوفة حالتين ولانها تتكامل
= مصفوفة الوحدة

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ -2 & 3 & -2 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 3 & -1 & 2 & 0 & 1 & 0 \\ -2 & 3 & -2 & 0 & 0 & 1 \end{array} \right]$$

$$R_2 = -3R_1 + R_2$$

$$R_3 = 2R_1 + R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & -7 & 2 & -3 & 1 & 0 \\ 0 & 7 & -2 & 2 & 0 & 1 \end{array} \right]$$

$$R_2 = -\frac{1}{7} R_2$$

Find the adjoint of matrix A

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$$

(i) Find the minor M_{ij} :

$$M_{11} = \begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix} = 2 - 0 = 2$$

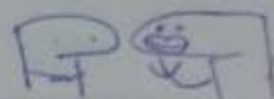
$$M_{12} = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 4 - 1 = 3$$

$$M_{13} = \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} = 0 - 1 = -1$$

$$M_{21} = \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} = 2 - 0 = 2$$

$$M_{22} = \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} = 2$$

Mazen



$$V_1 = (1, 2, 3) \quad V_2 = (2, -1, 1, 0)$$

$$\text{and } V_3 = (0, 5, -1, 2)$$

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & -1 & 5 \\ 0 & 1 & -1 \\ 1 & 0 & 2 \end{pmatrix}$$

→ RREF (درجه نوبت)

$$R_2 \leftarrow -2R_1 + R_2$$

$$R_4 \leftarrow -R_1 + R_4$$

dependent



$$\text{at } \lambda = 3 \quad \begin{pmatrix} 3-3 & 0 \\ 1 & 2-3 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$1x + (-1)y = 0$$

$$x - y = 0$$

$$x = y$$

~~$x = y$~~

$$\begin{pmatrix} x \\ x-y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$A = \begin{bmatrix} 1-\lambda & 2 & 2 \\ 2 & 3-\lambda & -2 \\ -5 & 3 & 8-\lambda \end{bmatrix} = 0$$

$$V = \begin{bmatrix} -5 \\ 11 \\ -7 \end{bmatrix}$$

$$u_1 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ 5 \\ 5 \end{bmatrix} \text{ and } u_3 = \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} -5 \\ 11 \\ -7 \end{bmatrix} = C_1 \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ 5 \\ 5 \end{bmatrix} + C_3 \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix}$$

$$C_1 + 2C_3 = -5$$

$$-2C_1 + 5C_2 = 11$$

$$2C_1 + 5C_2 + 8C_3 = -7$$

No solution

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بجبر

Vectors in P^2 is Linearly independent or Linearly dependent

$$S = \{1+x-2x^2, 2+5x-x^2, x+x^2\}$$

$$C_1(1+x-2x^2) + C_2(2+5x-x^2) + C_3(x+x^2) = 0$$

$$\begin{bmatrix} C_1 & C_2 & C_3 & | & 0 \\ -2 & -1 & 1 & | & 0 \\ 1 & 5 & 1 & | & 0 \\ 1 & 2 & 0 & | & 0 \end{bmatrix}$$

مساكن ال x^2 ال x ال 1

$$R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 5 & 1 & | & 0 \\ -2 & -1 & 1 & | & 0 \\ 1 & 2 & 0 & | & 0 \end{bmatrix}$$

$$\begin{aligned} R_2 &= 2R_1 + R_2 \\ R_3 &= -R_1 + R_3 \end{aligned}$$

$$\begin{bmatrix} 1 & 5 & 1 & | & 0 \\ 0 & 9 & 3 & | & 0 \\ 0 & -2 & -1 & | & 0 \end{bmatrix}$$

$$R_2 = \frac{1}{3} R_2$$

$$\begin{bmatrix} 1 & 5 & 1 & | & 0 \\ 0 & 1 & \frac{1}{3} & | & 0 \\ 0 & -2 & -1 & | & 0 \end{bmatrix}$$